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ChatEndoscopist: A Domain-Specific Chatbot with Images for Gastrointestinal Diseases

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Abstract. This study aims to enhance domain-specific medical knowledge within large language models (LLMs) by developing a chatbot, chatEndoscopist, a specialized model for oesophageal cancer. In particular, the chatbot incorporates related images to further elucidate the retrieved content while providing answers. Fine-tuned BioMistral LLM with 50 related documents, a dataset specifically curated for medical literature, ChatEndoscopist was compared to ChatGPT. For text answers, despite its specialized training, ChatGPT appears to outperform ChatEndoscopist in precision (0.210 vs. 0.148), recall (0.323 vs. 0.049), and F1 score (0.266 vs. 0.099). ChatGPT also demonstrated superior lexical diversity with a Type-Token Ratio (TTR) of 0.772 and Lexical Density of 0.813, compared to ChatEndoscopist's TTR of 0.717 and Lexical Density of 0.781. This in part, could be due to the limited documents to fine tune. However, the related images are mostly retrieval with regarding to user's queries. Future work will focus on incorporating more related papers to balance specialized accuracy with broader linguistic flexibility.

Keywords. large language model, LLaMA, ChatGPT, oesophageal cancer, chatbot

1. Introduction

Recent advances in artificial intelligence (AI), particularly large language models (LLMs), have led to the development of powerful chatbots that are capable of performing human-like conversations. Since the release of Chat Generative Pre-trained Transformer (ChatGPT) by OpenAI in 2022 [1], chatbots have demonstrated significant potential in tasks such as writing, translation, and even coding, as well as applications in healthcare [2]. While general-purpose AI models like ChatGPT perform well across a variety of tasks, they often lack of domain-specific medical knowledge [3], required for accurate responses in specialized areas like oesophageal cancer and endoscopy.

Gastrointestinal tract (GI) cancers, including those of the oesophagus, stomach, bowel, and pancreas, accounted for 26.3% of global cancer cases and 35.4% of cancer-

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related deaths in 2018 [4]. As of 2020, GI cancers were the second leading cause of cancer deaths worldwide, after lung cancer, with a five-year survival rate below 20% [4]. For example, oesophageal cancer, particularly oesophageal adenocarcinoma, is often diagnosed at advanced stages, leading to poor patient outcomes [5]. Early detection and educating healthcare professionals and patients on risk factors, symptoms, and diagnostic tools are essential [6].

The detection of GI diseases is usually performed by endoscopy, which involves inserting a tiny camera into the oesophagus, stomach or intestines or by swallowing a capsule-sized camera, mainly for small intestines. Hence the accompanying images related to GI diseases are more understandable to ordinary users than text-only answers by explaining context clearer. At present, while most AI-driven chatbots have emerged as useful tools for providing accurate, up-to-date medical information, supporting education and clinical decision-making [3], they are mainly text-based. Examples specific to GI diseases include chatbots designed to manage gastroesophageal reflux disease, colonoscopy screening guidelines, and clinical documentation [7]. AI chatbots have also been applied in oncology, aiding early detection and precision diagnostics, which is critical to patient outcomes [8].

In this study, ChatEndoscopist is developed not only as a domain-specific AI model, but also with the retrieval of related images illustrating the text answers, with a particular focus on oesophageal cancer. By integrating specialized medical knowledge, trained through published related papers, ChatEndoscopist aims to offer more accurate, contextually relevant responses than general query platforms (e.g. Google or Wikipedia), leading to providing more informative educational information with the help of endoscopic images for both the public and patients.

2. Methods

Figure 1 schematically illustrates the workflow of ChatEndoscopist.

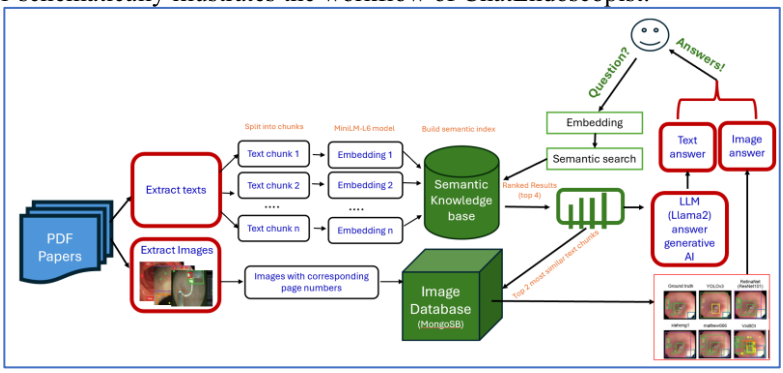


Figure 1. The work flow diagram of ChatEndoscopist, including chucking, embedding and retrieval.

Firstly, a collection of published papers and documents related to GI tract diseases are inputted into the system. Then content extraction takes place, which split each document into chunks. In LLM, chucking is an essential technique that divides large documents or texts into smaller pieces called chucks. This process will enable language models to process long texts efficiently while preserving context and meaning. In essence, a chuck refers to a ‘unit of information’ and is expected to present a complete ‘idea’. Hence, a

document is a collection of unified concepts and is represented by 'pieces of units of text or token'. In LLMs, a token is a basic unit of text that can be a word, character, or a sequence of characters. This is essential because LLMs have token limits, which restrict the number of text LLMs can process at each time. For example, ChatGPT-3 has a token limit of 4096, including both input and output tokens.

In this study, each chunk contains 1000 characters with 200 characters overlapping. Chunked texts are subsequently converted (embedded) into numeric values through the application of a sentence-transformers model MiniLM-L6[2], which maps sentences and paragraphs to a 384-dimensional dense vector space to allow clustering or semantic search. The resulted vectors are then stored in a Faiss Vector Database, the *Facebook AI Similarity Search*, which is a library allowing search of multimedia documents in a speeding way by finding similar contents based on the closer distance to a query in a vector space. In addition, images are extracted as well with corresponding page numbers to associate with text chunks and are stored in a database of MongoDB.

Upon deployment of this chatbot, the user's questions are transformed into an embedding model of MiniLM-L6. Then the top 4 most similar text chunks observed in the vector space are determined empirically. Together with the conversation history, these text chunks are sent to the pretrained language model BioMistral-7B to generate answers. By employing Mistral as its foundation model, BioMistral is further trained on PubMed Central papers and comprise a collection of open-source pretrained LLM tailored for medical domain [9]. For image retrieval, the top 2 most similar text chunks are applied to find the corresponding image page number and thereafter images from MongoDB, which is to ensure both the related images being retrieved back and less relevant images being avoided. To specialize the chatbot in oesophageal cancer and endoscopy, BioMistral model is further fine-tuned using a dataset of collected 50 research papers specifically focused on these topics. The chatbot is using Streamlit[3], a Python-based open-source framework to build web applications. The maximum sequence length is 512 tokens with max batch size of 8 and epoch number of 5.

Three specific prompts related to oesophageal cancer were used to evaluate the key aspects such as aetiology, risk factors, and disease progression: 1) *"What is Barrett's Oesophagus"*; 2) "*What are the risk factors for Barrett's oesophagus and oesophageal cancer"*; and 3) *"How does Barrett's oesophagus progress to cancer."* Each prompt was crafted to assess the chatbot's ability to retrieve, synthesize, and present clinically relevant information.

Metrics of BERTScore [10] and Lexical Diversity [11] were employed to evaluate the performance of chatEndoscopist compared to ChatGPT-4.0. BERTScore is used to assess the semantic similarity between the chatbot responses and reference answers, applying the `bert-score` Python library (version 0.3.13) with the `roberta-large` model. Responses and references are tokenized and mapped to dense embedding whereas cosine similarity is employed to match tokens and calculate Precision, Recall, and F1 Score for each prompt, with averages being reported. Lexical diversity is measured via: 1) *Type-Token Ratio (TTR)*: that measures the ratio of unique words (types) to the total number of words (tokens), reflecting vocabulary variety; 2) *Lexical Density*: that measures the proportion of content-heavy words (nouns, verbs, adjectives, and adverbs) in the generated responses. Python programs are used to calculate these metrics.

[2] https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2.
[3] https://streamlit.io/.

3. Results and Discussion

Figure 2 demonstrates the chatbot's capability to handle specialized medical inquiries, showing its response and associated image retrieval functionality for a 'specularity artefact' query by providing relevant endoscopic image outputs. Figure 3 compares the responses of ChatEndoscopist and ChatGPT to a query about Barrett's Oesophagus. ChatEndoscopist delivers a detailed and clinically oriented response, highlighting tissue involvement and disease progression, which is ideal for medical professionals. In contrast, ChatGPT offers a broader, educational overview, that is more suitable for general knowledge and patient education.

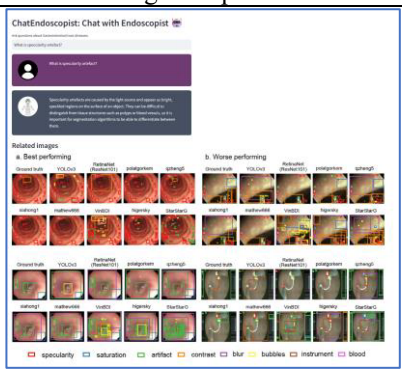


Figure 2. ChatEndoscopist Textual Response and Associated Image Retrieval on 'specularity artefact' Query.

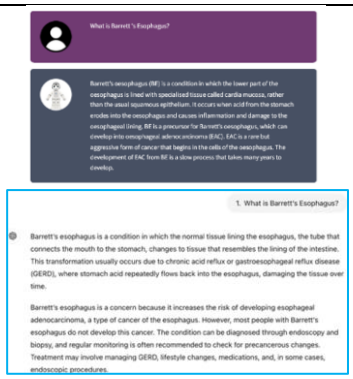


Figure 3. Responses from ChatEndoscopist (top) vs ChatGPT (bottom) on Barrett's Oesophagus.

To quantitatively assess the performance of chatEndoscopist and ChatGPT, both models have been evaluated on three key prompts related to oesophageal cancer using two metrics of BERTScore (measuring semantic similarity) and Lexical Diversity (assessing vocabulary richness). Table 1 summarizes the overall performance.

Table 1. Performance Comparison Between ChatEndoscopist and ChatGPT

Item	BERTScore Performance			Lexical Diversity	
	Precision	Recall	F1Score	Type-Token Ratio (TTR) (Mean, CI)	Lexical Density (Mean, CI)
ChatEndoscopist	0.148	0.049	0.099	0.717 (0.512 - 0.921)	0.781 (0.731 - 0.831)
ChatGPT 4.0	0.210	0.323	0.266	0.772 (0.727 - 0.818)	0.813 (0.717 - 0.910)

The BERTScore results show that ChatGPT outperforms ChatEndoscopist across all metrics in semantic similarity, achieving an F1 Score of 0.266 compared to 0.099 for chatEndoscopist. This indicates that ChatGPT's responses align more closely with the reference answers, while chatEndoscopist, through providing medically accurate information, focuses on domain-specific terminology. In terms of Lexical Diversity, ChatGPT also demonstrates superior performance, achieving a TTR of 0.772 (95% CI: 0.727–0.818) and Lexical Density of 0.813 (95% CI: 0.717-0.910), reflecting a more varied vocabulary and content-heavy responses. Meanwhile, chatEndoscopist scored a TTR of 0.717 (95% CI: 0.512-0.921) and Lexical Density of 0.781 (95% CI: 0.731-0.831), showing a more focused and repetitive use of specialized medical terminology. Wider CIs for ChatEndoscopist's TTR highlight variability in vocabulary diversity across prompts, while narrower CIs for Lexical Density indicate stable, information-rich responses. Conversely, ChatGPT's narrower TTR CIs suggest consistent vocabulary

variety, while its wider Lexical Density CIs reflect variability in content density. On the other hand, for images, most of related images are retrieved based on the queries.

4. Conclusions

This study investigates the feasibility of developing a domain-specific chatbot with the retrieval of related accompanying images. Initial results have demonstrated images clearly helping the understanding of domain knowledge, especially when text answers are not comprehensive. However, in comparison with the existing LLM of ChatGPT, the text answers appear to be less accuracy when measured using BERTScore and Lexical Diversity. Further studies will be conducted to evaluate medical experts' view on the retrieved answers. In addition, the publications to be used for fine-tuning in this study will be increased from our current 50 papers. Studies have shown that models fine-tuned with sufficient domain-specific knowledge can outperform general-purpose large language models (LLMs) when applied to specialized topics. Future work includes bringing expert endoscopists onboard to further refine retrieved answers.

Acknowledgements
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